

Robotic Planning: Evaluating Futures Through Expected Free Energy

Executive Summary

Planning is the process by which a robot evaluates possible future action sequences and selects the one that best achieves its goals while managing uncertainty. In Active Inference, planning is formalized as the evaluation of policies (sequences of actions) according to their expected free energy — a quantity that combines the pragmatic value of reaching desired states with the epistemic value of reducing uncertainty and the risk of visiting undesirable states. This module examines how robotic planning algorithms — from motion planning through configuration space to task planning through symbolic state spaces — implement this evaluation, and how the depth and breadth of planning are constrained by computational resources and real-time deadlines.

Learning Objectives

1. Define robotic planning as the evaluation and selection of action sequences that minimize expected free energy over future time horizons.
2. Analyze motion planning algorithms (RRT, PRM, A*) as search procedures over the robot's generative model of configuration space.
3. Distinguish between motion planning (geometric), trajectory planning (kinodynamic), and task planning (symbolic) as different levels of the planning hierarchy.
4. Evaluate the trade-offs between planning depth, computational cost, and plan quality in real-time robotic systems.
5. Connect robotic planning to Active Inference's expected free energy decomposition into pragmatic, epistemic, and risk components.

Introduction

The previous module on Communication examined how robots exchange information with other agents. This final module in the Robotic Systems unit addresses the most cognitively

demanding capability: planning — the ability to reason about the future and select actions that lead to desired outcomes over extended time horizons. Planning integrates everything covered in the preceding modules: the system architecture that defines what is possible (Module 1), the agent framework that provides goals and preferences (Module 2), the perceptual processes that estimate the current state (Module 3), the cognitive representations that support temporal reasoning (Module 4), the action capabilities that constrain what can be executed (Module 5), the learned models that predict outcomes (Module 6), and the communication channels that provide external constraints and goals (Module 7). Planning is where it all comes together.

Key Concepts

1. Planning as Expected Free Energy Minimization

In Active Inference, planning is the selection of a policy — a sequence of future actions — that minimizes expected free energy. Expected free energy for a policy decomposes into three components:

Pragmatic value: the expected proximity of future observations to the agent's preferred observations. A mobile robot planning a path to its goal location evaluates pragmatic value by how close each candidate path brings it to the goal. A manipulator planning a grasp evaluates pragmatic value by how likely each grasp candidate is to achieve a stable hold.

Epistemic value: the expected reduction in uncertainty about hidden states. A mobile robot approaching an unexplored area gains epistemic value by planning paths that reveal new regions of the map. A manipulator gains epistemic value by planning viewpoints that resolve uncertainty about object poses before committing to a grasp.

Risk: the expected probability of visiting states that are far from the agent's preferred states. A mobile robot loses expected free energy by planning paths that pass too close to obstacles (risk of collision). A manipulator loses by planning trajectories that approach joint limits or singularities (risk of loss of control).

Classical robotic planning methods typically optimize only pragmatic value (shortest path to goal) with risk handled as hard constraints (collision avoidance). Active Inference provides a unified framework that naturally balances all three components, producing plans that are goal-

directed, information-seeking, and cautious — exactly the properties desired in autonomous robotic systems.

2. Motion Planning Through Configuration Space

Motion planning — finding a collision-free path from a start configuration to a goal configuration — is one of the most studied problems in robotics. The configuration space (C-space) of a robot is the set of all possible joint configurations, and obstacles in the workspace project into forbidden regions in C-space. The planning problem is to find a continuous path through the free region of C-space.

Sampling-based planners like RRT (Rapidly-exploring Random Trees) and PRM (Probabilistic Roadmap) explore C-space by randomly sampling configurations and connecting them with collision-free edges. RRT grows a tree from the start configuration by repeatedly sampling a random configuration and extending the nearest tree node toward it. RRT* improves on RRT by rewiring the tree to find asymptotically optimal paths. These planners are probabilistically complete — they will find a path if one exists, given enough time — but they do not guarantee optimality in finite time.

From an Active Inference perspective, sampling-based planners are performing stochastic search over the generative model's prediction of future configurations. Each sample tests a hypothesis about whether a particular configuration is collision-free (evaluating the observation model against the obstacle map). The tree structure represents the set of reachable configurations, and path selection from start to goal evaluates pragmatic value (reaching the goal) subject to risk constraints (avoiding collisions).

Search-based planners like A* *operate on discretized representations of C-space or workspace*. A* finds the lowest-cost path through a graph by combining the cost-so-far with a heuristic estimate of the cost-to-go. For mobile robot navigation on a 2D costmap, A* finds the shortest (or lowest-cost) path from the robot's current position to the goal, with costs encoding obstacle proximity and terrain difficulty. The heuristic function — typically Euclidean distance to the goal — provides an optimistic estimate that guides search toward the goal, corresponding to the pragmatic component of expected free energy.

3. Trajectory Planning Under Dynamic Constraints

Motion planning finds geometric paths; trajectory planning adds timing — specifying not just where the robot goes but when it arrives at each waypoint. This is essential for robots operating in dynamic environments with moving obstacles, coordination requirements, or physical constraints on velocity and acceleration.

For a UR5 manipulator, trajectory planning converts a geometric joint-space path into a time-parameterized trajectory that respects velocity limits (maximum 180 degrees per second per joint), acceleration limits, and jerk limits (for smooth motion). The time-optimal trajectory minimizes execution time while satisfying all constraints. In Active Inference terms, the trajectory specifies the temporal sequence of preferred proprioceptive states, and the timing constraints ensure that these preferences are physically achievable given the actuator limitations.

For a mobile robot operating among pedestrians, trajectory planning must predict the future positions of moving agents and plan paths that avoid collisions not just with static obstacles but with predicted future positions of dynamic obstacles. The Dynamic Window Approach (DWA) and Timed Elastic Band (TEB) planners generate kinodynamically feasible trajectories that account for the robot's velocity constraints and the predicted motion of nearby agents. This is planning under a generative model that includes the dynamics of other agents — a richer model that enables coordination with a dynamic environment.

4. Task Planning at the Symbolic Level

Task planning operates at a higher level of abstraction than motion planning, reasoning about sequences of discrete actions rather than continuous motions. A task planner for a manipulation robot might reason about sequences like: locate bolt, pick bolt, move to assembly, insert bolt, tighten bolt. Each action is described by its preconditions (what must be true for the action to be applicable) and effects (what becomes true after the action executes).

PDDL (Planning Domain Definition Language) is the standard formalism for task planning in robotics. A PDDL problem specifies the initial state (what objects exist and their properties), the goal state (what properties must be achieved), and the available actions. Classical planners

like Fast Downward and LAMA find action sequences that transform the initial state into a state satisfying the goal.

In Active Inference terms, task planning corresponds to evaluating policies over a discrete state-space generative model with long time horizons. Each policy is a sequence of symbolic actions, and expected free energy evaluation predicts the sequence of state transitions and observations that each policy would produce. The planner selects the policy with the lowest expected free energy — typically the one that reaches the goal state (high pragmatic value) with the fewest actions (low complexity cost) and minimal risk of failure.

5. Hierarchical Planning and Anytime Computation

Real robotic systems combine task planning, motion planning, and trajectory planning in a hierarchy. The task planner selects what to do (symbolic action sequence), the motion planner selects how to move (geometric path), and the trajectory planner selects when to be where (timed trajectory). This hierarchy enables tractable computation by decomposing a complex planning problem into manageable subproblems.

Anytime planning algorithms provide an initial feasible solution quickly and then progressively improve it as more computation time is available. ARA (*Anytime Repairing A*) finds an initial suboptimal path rapidly and then tightens the suboptimality bound with continued computation. RRT* finds an initial feasible path quickly and asymptotically approaches the optimal path with continued sampling. These anytime properties are essential for real-time robotic operation, where a feasible plan that meets the control deadline is infinitely better than an optimal plan that arrives too late.

In Active Inference terms, anytime planning corresponds to approximate inference with a time-varying precision bound. Initially, the agent settles for a coarse posterior over policies (the first feasible plan). As time permits, it refines this posterior, reducing the approximation error and potentially switching to a better policy. The expected free energy of the best-so-far policy provides a monotonically improving bound on the value of continued computation, allowing the robot to decide when planning is "good enough" and execution should begin.

Active Inference Connection

Planning in Active Inference is the prospective evaluation of policies — imagining the sensory consequences of action sequences and selecting the sequence that minimizes expected free energy. This process is fundamentally model-based: the robot simulates its generative model forward in time under each candidate policy, evaluating the predicted observations against its preferences. The depth of planning (how many time steps ahead the model is unrolled) trades off against computational cost. Deep planning enables sophisticated, multi-step task completion; shallow planning supports fast, reactive responses. The unified expected free energy objective ensures that planning naturally balances goal achievement, information seeking, and risk avoidance without requiring separate objective functions for each.

Applications

Case Study 1: RRT*-Based Manipulation Planning for a UR5

A UR5 manipulator must plan a collision-free trajectory to reach a grasp pose inside a cluttered shelf. The 6-DOF configuration space is vast, and the shelf's narrow opening creates a constrained passage that the arm must thread through. An RRT* planner samples random configurations, checks each for collision against a 3D model of the shelf (evaluating the observation model: does this configuration overlap with known obstacles?), and grows a tree of feasible configurations from the start pose. Within 200 ms, the planner finds an initial feasible path. Over the next 500 ms, it continues sampling and rewiring, finding a shorter path that avoids unnecessary detours. The resulting trajectory is then smoothed and time-parameterized using a minimum-jerk profile that respects joint velocity and acceleration limits. The total planning time of 700 ms fits within the robot's 1-second planning budget, allowing execution to begin promptly. The pragmatic value (reaching the grasp pose), risk (avoiding shelf collisions), and computational constraint (1-second deadline) are all satisfied by this hierarchical, anytime planning approach.

Case Study 2: Behavior Tree Task Planning for Mobile Robot Delivery

A Clearpath Jackal robot performing package delivery in an office building uses a behavior tree as its task-level planner. The tree encodes a hierarchical plan: check if a delivery request exists, navigate to the pickup location, wait for loading, navigate to the delivery location, wait for unloading, return to base. Each navigation subtask invokes the Nav2 action server, which handles motion planning and trajectory execution. The behavior tree implements reactive replanning: if navigation fails (obstacle blocking the path, elevator out of service), the tree's fallback nodes trigger recovery behaviors (spin, wait, try alternate route) before re-attempting. This hierarchical structure — symbolic task planning through the behavior tree, geometric motion planning through Nav2, and trajectory execution through the local controller — demonstrates the three planning levels operating in concert. The expected free energy of each behavior tree branch is implicitly evaluated through the tree's condition checks and priority ordering, with the highest-priority feasible branch selected for execution.

Cross-References

- **Module 1 (Systems):** The system architecture that defines the planning problem's state space and constraints
- **Module 2 (Agents):** The agent's preference model that defines planning objectives
- **Module 4 (Cognition):** The generative model that planning simulates forward in time
- **Module 5 (Action):** The action capabilities that constrain what plans are executable

Summary

Concept	Definition	Robotics Example
Expected Free Energy	Objective function combining pragmatic, epistemic, and risk components	Evaluating a navigation path for goal proximity, information gain, and collision risk
Motion Planning	Finding collision-free paths through configuration space	RRT* generating a joint-space path for a UR5 in a cluttered shelf
Trajectory Planning	Adding timing constraints to geometric paths	Minimum-jerk time parameterization of a manipulator path
Task Planning	Selecting sequences of symbolic actions to achieve goals	PDDL planner generating a pick-place-insert sequence
Anytime Planning	Providing improving solutions as computation time increases	RRT* progressively shortening a path while the robot waits
Hierarchical Planning	Decomposing planning across abstraction levels	Behavior tree (task) + Nav2 planner (motion) + DWB (trajectory)

References

1. Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.
2. LaValle, S. M. (2006). *Planning Algorithms*. Cambridge University Press.
3. Karaman, S., & Frazzoli, E. (2011). Sampling-based algorithms for optimal motion planning. *International Journal of Robotics Research*, 30(7), 846-894.
4. Colledanchise, M., & Ogren, P. (2018). *Behavior Trees in Robotics and AI*. CRC Press.
5. Fountas, Z., et al. (2020). Deep active inference agents using Monte-Carlo methods. *NeurIPS*.